Basic Structure (Boilerplate)

Every program has an "Entry Point" where execution begins

C++

```cpp
#include <iostream>
using namespace std;

// Entry Point
int main() {
    cout << "Hello";
    return 0;
}
```

Java

```java
public class Main {
    // Entry Point
    public static void main(String[]
        args) {
        System.out.println("Hello");
    }
}
```

Python

```python
# No boilerplate needed!
# Script executes top-to-bottom

print("Hello")
```

Note: C++ and Java require a specific function ('main') to start.

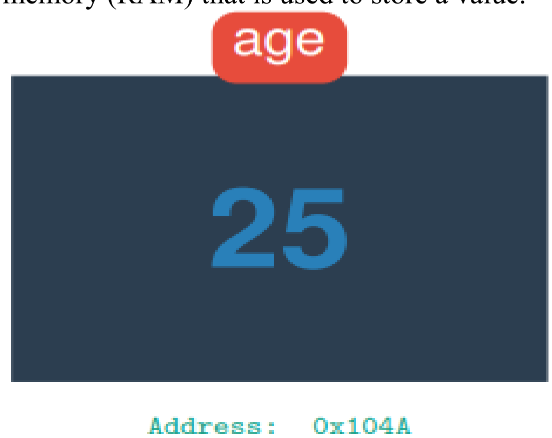
What is a Variable?

Definition:

A **Variable** is a named storage location in the computer's memory (RAM) that is used to store a value.

Box Analogy (Easy Understanding):

- **Label Name** → Variable Name (e.g., `age`)
- **Contents** → Stored Value (e.g., 25)
- **Box Size** → Data Type (e.g., Integer)
- **Address** → Memory Location (e.g., 0x104A)



Identifiers (Naming Variables)

Definition:

An **Identifier** is the name given to a variable, function, or class.

The 3 Golden Rules

1. Can contain **Letters** (A-Z, a-z), **Digits** (0-9), and **Underscores** (_).
2. Must **NOT** start with a digit. (e.g., `1stPlace` is illegal).
3. Must **NOT** be a Keyword (reserved word like `int`, `class`, `if`).

Valid	Invalid	Reason
`studentAge`	`student Age`	Contains space
`grade_1`	`1stGrade`	Starts with number
`_total`	`float`	Reserved keyword
